

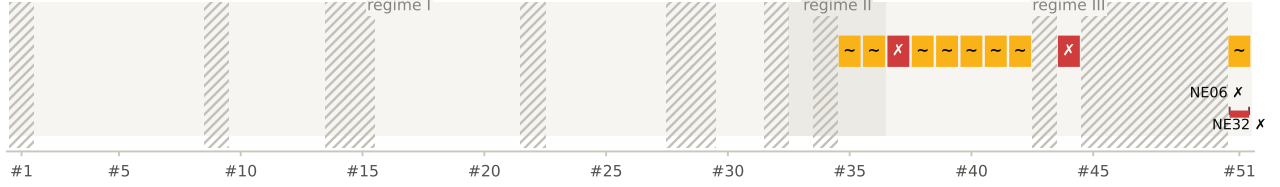
H13: Model families carry their own fields, and couple by family

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1.5/5 Usefulness 2/5

Bottom line. Model families differ in how they write, faintly in what they do, and not in whom they follow; rooms, not vendors, set the coupling.

level: hypothesis · pipeline checks 86% · updated 2026-10-04

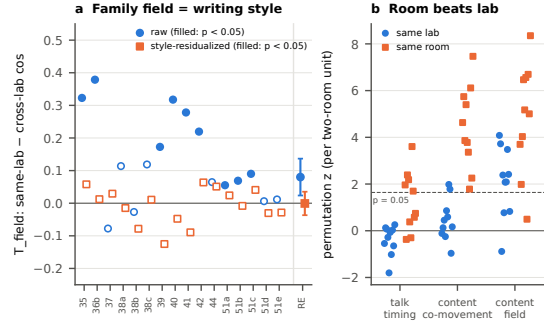


Goal periods tested (4 failed, 8 mixed). ✓ supported ✗ failed ~ mixed d descriptive ... pending.

Heavy outline: confirmatory test on the locked holdout. Hatched columns: held-out goal periods. Brackets: tests spanning a natural experiment.

WHAT HAPPENED

The data is 15 counted non-holdout units (#35–#44, #51; $N = 10\text{--}27$). The state is each agent’s daily content vector. After removal of each day’s common field, same-lab agents share a content direction in 9/15 units: random-effects $T = 0.08$ [0.02, 0.14]. The direction is stable across periods (split-half 0.88 against 0.76, $p = 0.008$). **It is writing style:** it survives style residualization in 0/9 units. Talk timing and content co-movement show **no family coupling; rooms carry coupling**. Round 1b (second embedding model, DQ5 style residuals, dedupe, corrected talk table) reproduces every number, and the talk null is 2.3× tighter. The new behavioral test (Jev v3 states, real failures, commit and write rates) **fails** its rule: 3/15 units, RE $T_B = 0.075$ [0.03, 0.12]. That is a weak trace of work habits. Natives: NE06 (a Google-only change) is village-wide in the logs, newcomers match by behavior in 3/9 ($p = 0.14$), and #35 is mixed. The holdout has not run.



Observables. (a) The family-field statistic T_{field} per unit before (blue) and after (orange) regression on writing-style features. Filled: lab-permutation $p < 0.05$. RE: random-effects summary with 95% CI. The field vanishes under style control. (b) Two-room units: permutation z of the same-lab and same-room pair coefficients. Talk timing and content co-movement follow rooms, not labs. The static content field has a lab term too, which is style.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

PHYSICAL MODEL

Model 11 (vector spins) plus models 01/02 in K -block mean field. $v_{i,d}$ is agent i ’s mean whitened chat embedding on day d . g_d is the day (goal) field, estimated by the day mean m_d . h_f is the field of lab $f(i)$, and a_i is an agent offset. The fluctuations $x_{i,d}$ couple through a $K \times K$ family matrix ($K = 2\text{--}5$ multi-member labs). ε is sampling noise, and H_i is the agent field. C is the talk-spin or co-movement correlation matrix. y_{ij} is a pair outcome, and $r(i)$ is the agent’s room.

$$\begin{aligned} v_{i,d} &= g_d + h_{f(i)} + a_i + x_{i,d} + \varepsilon_{i,d}, & H_i &= \langle v_{i,d} - m_d \rangle_d \\ T_{\text{field}} &= \langle \cos(H_i, H_j) \rangle_{f(i)=f(j)} - \langle \cos(H_i, H_j) \rangle_{f(i) \neq f(j)} \\ J &\approx -C^{-1} \text{ (off-diag.)}, & J_{ij} &= J_{f(i)f(j)}, & \Delta &= J_{\text{in}} - J_{\text{out}} \\ y_{ij} &= c + b_{\text{lab}}[f(i) = f(j)] + b_{\text{room}}[r(i) = r(j)] \end{aligned}$$

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RESULTS VS PREDICTIONS

FAITHFULNESS SCORECARD

Prediction	Round 1 → round 1b (bge; gte)	Verdict
P1 family field	9/15, RE 0.080 → 9/15, 0.080; gte mixed 8/15, 0.085	
P2 not only style	0/9 → 0/9 (own S-a); 1/9, 0/8 (DQ5 × residuals)	
P3 family invariance	0.88 vs 0.76 → 0.88 / 0.90 vs 0.76 / ✓ 0.74	
P5 no talk homophily	RE 0.002 [−0.04, 0.05] → −0.003 ✓ [−0.02, 0.02]	
P6 no content ho-mophily	1/15 → 1/15 (both models)	✓
P7 rooms carry coupling	talk 8 → 10/10; co-movement 9/10, ✓ gte 10/10	
B1 behavioral field (HH267)	3/15; RE T_B 0.075 [0.03, 0.12]	×
B2 behavior beyond style	4/15; RE 0.038 [−0.01, 0.09]	×
NE06 / NE32 / G35 (native)	village-wide change / 3 of 9 / words follow rooms / ~	×

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	0
D unfitted predictions	1	I transfer	1
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

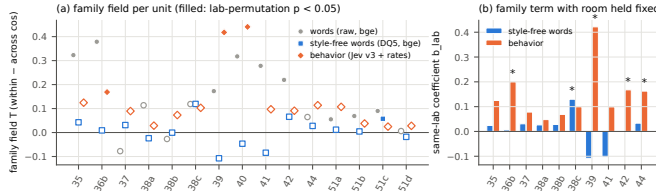
CAVEATS

The behavioral field can be harness, not disposition. It lives in file-write and commit cadence, which follow each provider’s tool conventions. NE06 cannot separate the two, because the change hits every agent. Jev’s labels read window evidence that can include an agent’s narration. In #51 roles dominate behavior, so same-role pairs are dropped. The no-coupling results exclude only couplings $\gtrsim 0.1$. The holdout (C1–C6) is untouched.

ROUND 2

Test the behavioral field against a harness covariate: tool-call conventions per provider. Use NE20 (Anthropic one call per turn, holdout) only if Vivian approves the confirmatory run.

- **Essence**. Do vendors differ in decisions, not prose?
- **H13-R1** Policy fingerprint: P(next behavior state — current state, context features) differs by family beyond agent identity.
- **H13-R2** Families differ in susceptibility: who listens to whom (HH98).
- **H13-R3** Mixed-vendor periods out-produce single-vendor ones (descriptive, confounded; work ledger).



Second figure. (a) The family field per unit in three channels: raw words (gray), words after DQ5 style residualization (blue) and behavior (orange). Filled: $p < 0.05$. Style-free words carry nothing. Behavior carries a little, mostly in the build weeks #39–#40. (b) Two-room units: the same-lab coefficient with the room held fixed. Behavior keeps a family term (RE 0.14); style-free words do not.

GOAL PERIODS IN DETAIL

G35 ~ 1b: mixed (round 1: mixed; same in both embedding mod...
G36 ~ 1b: mixed (round 1: mixed; same in both embedding mod...
G37 × 1b: failed (round 1: failed; same in both embedding m...
G38 ~ 1b: mixed (round 1: mixed; same in both embedding mod...
G39 ~ 1b: mixed (round 1: mixed; same in both embedding mod...
G40 ~ 1b: mixed (round 1: mixed; same in both embedding mod...
G41 ~ 1b: mixed (round 1: mixed; same in both embedding mod...
G42 ~ 1b: mixed (round 1: mixed; same in both embedding mod...
G44 × 1b: failed (round 1: failed; same in both embedding m...
G51 ~ 1b: mixed (round 1: mixed; same in both embedding mod...

+2 more periods: see the card.

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